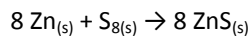


Stoichiometry - Limiting & Excess

- 1) In a reaction to produce zinc sulfide, 5.00 g of zinc and 3.00 g of sulfur, S₈, are available. The balanced equation is given below:

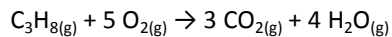


- (a) Identify the limiting reagent.

- (b) Calculate the mass of zinc sulfide formed.

(7.45g)

- 2) Propane, C₃H_{8(g)}, used in gas barbecues, burns in oxygen to produce carbon dioxide and water according to the following equation:



In a reaction, 100.7 g of propane and 367.4 g of oxygen gas are available.

- (a) Identify the limiting reagent in this reaction.

- (b) Calculate the mass of water formed.

(164.6g)

- 3) One of the reactions in the formation of acid rain is the synthesis reaction of sulfur dioxide, $\text{SO}_{2(g)}$, with $\text{O}_{2(g)}$ to form sulfur trioxide, $\text{SO}_{3(g)}$, as the only product.

(a) Write a balanced equation for this reaction.

(b) If 4.55 kg of sulfur dioxide and 2.88 kg of oxygen gas are available, what mass of sulfur trioxide will be produced?

(5.69 kg)